

Bike Sales Performance Analysis

Tableau & Power BI | Sales Performance • Profitability • Product • Customer • Geographic Analysis

1. Project Overview

This project analyses bike sales transaction data from 2011–2016 to evaluate revenue performance, profitability, product performance, customer segments, and geographic markets. The analysis turns raw sales data into interactive dashboards and practical business insights.

Business Objective

Identify the major drivers of revenue and profit, understand differences in product and market performance, and highlight opportunities for revenue growth and margin improvement.

Dataset

113K+ transaction records covering January 2011 through July 2016. Key fields include Revenue, Profit, Cost, Order Quantity, Product, Product Category, Sub-Category, Country, State, Year, Month, Age Group, and Customer Gender.

Data note: The dataset does not contain a unique customer ID or unique order ID. Customer-related analysis therefore uses the available demographic fields and should not be interpreted as unique-customer counts.

Tools & Methodology

Excel was used for data inspection and validation. Tableau was used for exploratory analysis and visual storytelling. Power BI was used for executive reporting and interactive analysis, supported by DAX measures and Power Query transformations.

Core KPIs

Metric	Result
Total Revenue	\$85.27M
Total Profit	\$32.22M
Total Cost	\$53.05M
Total Units	1.35M
Profit Margin	37.8%

Business Questions

- How are revenue and profit performing over time?
- Which categories and sub-categories drive revenue and profitability?
- Which products combine strong sales and margins?
- Which geographic markets generate scale versus stronger margins?
- Which age groups contribute most revenue, and how balanced is revenue by gender?
- Where are the clearest opportunities for margin improvement?

2. Key Findings & Business Recommendations

Product & Category Performance

Bikes generated approximately **72.5% of total revenue**, making them the primary revenue engine. Accessories generated approximately **58.6% profit margin**, making them the strongest-margin category. This creates a clear distinction between the category driving scale and the category driving margin.

Road Bikes generated the highest revenue among bike sub-categories, while Mountain Bikes achieved a stronger margin despite lower revenue. Helmets were also among the strongest-margin product groups.

Geographic Performance

The **United States** generated the highest revenue at approximately \$28.0M. **Canada** generated the highest country-level profit margin at approximately 46.8%, despite being a smaller revenue market. This demonstrates why market evaluation should consider both revenue scale and profitability.

Customer Demographics

Adults aged **35–64** contributed approximately **49.9% of total revenue**. Revenue was relatively balanced between male and female customers, with male revenue at approximately \$43.3M and female revenue at approximately \$41.9M.

Revenue Trend

Because 2016 contains data only through July, it should not be compared directly with full-year 2015. On a like-for-like January–July basis, 2016 revenue was approximately **25.2% higher than January–July 2014**, while profit was approximately **20.0% higher**. The corresponding margin was about 1.7 percentage points lower.

Business Recommendations

1. **Use Bikes as the primary volume driver** while monitoring profitability by sub-category and product.
2. **Increase accessory cross-selling and bundling** to take advantage of higher accessory margins.
3. **Review lower-margin clothing products** for pricing, cost, and product-mix opportunities.
4. **Investigate Road Bike profitability** to understand why revenue leadership does not translate into the strongest margin.
5. **Study high-margin markets such as Canada** for pricing, product-mix, or customer factors that may provide useful lessons elsewhere.
6. **Monitor revenue and margin together** so growth is evaluated alongside profitability.

Project Outcome

The project demonstrates practical experience in data cleaning and validation, KPI development, sales and profitability analysis, product and customer segmentation, geographic analysis, interactive dashboard development, Tableau visual storytelling, Power BI reporting, DAX calculations, and business-focused insight generation.

Project Note

This project originated from a data analytics training course and was adapted and expanded for my portfolio through additional analysis, dashboard design, KPIs, visualisation, and business insights. The dashboards and analysis are presented as a portfolio project to demonstrate practical data analytics and business intelligence skills.